



SEMESTER END / BACKLOG SUBJECT EXAMINATIONS - JULY 2024

Program	: B.E. - Common to all Programs	Semester	: I / II
Course Name	: Introduction to Civil Engineering	Max. Marks	: 100
Course Code	: ESC131 / ESC231	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Discuss on the following structural elements: CO1 (06)
(i) Column (ii) Beam (iii) Slab.
 - Mention the applications of (i) Reinforced Cement Concrete (ii) Plain Cement Concrete. CO1 (07)
 - Elaborate briefly on the roles and responsibility of (i) Transportation Engineer (ii) Geotechnical Engineer. CO1 (07)
- Mention the functions of the following structural elements: CO1 (06)
(i) Plinth (ii) Masonry wall.
 - List out the properties of cement and mortar used in the construction. CO1 (07)
 - Discuss the scopes of the following fields of Civil Engineering CO1 (07)
(i) Construction planning and management. (ii) Surveying.

UNIT - II

- Briefly describe sustainable development goals. CO2 (10)
 - Explain smart city concept and describe the features of smart city concept. CO2 (10)
- Explain: CO2 (10)
i) Energy efficient buildings
ii) Temperature and Sound control in buildings.
 - What is solid waste management? What are the functional elements of solid waste management? CO2 (10)

UNIT - III

- State and Prove Varignons Theorem. CO3 (05)
 - With relevant sketches explain support reactions. CO3 (05)
 - Determine the Reactions at the supports for the beam shown in Fig. 5(c). CO3 (10)

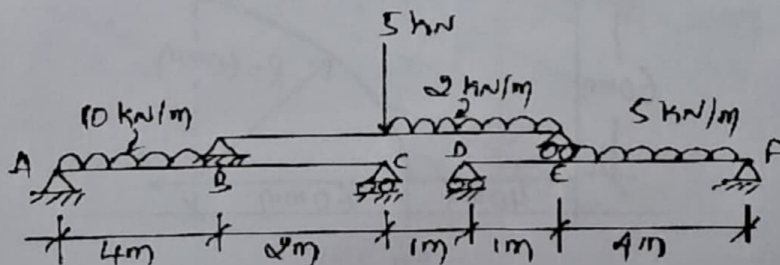


Fig. 5(c)

ESC131/ESC231

6. a) What is Free Body diagram? Write the conditions of equilibriums for coplanar force systems. CO3 (05)
 b) Find the moment of system of forces about point "A" as shown in the Fig. 6(b). CO3 (05)

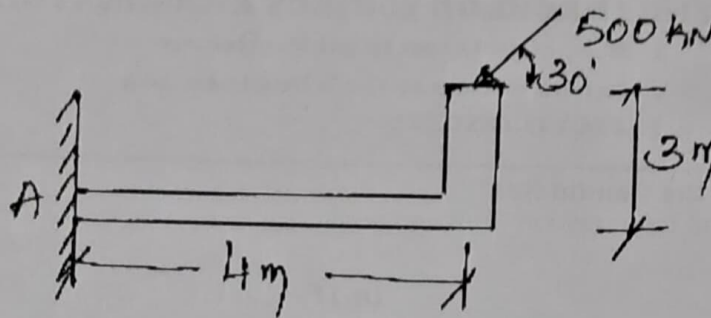


Fig. 6(b)

- c) Determine the forces in the various members for the system shown in Fig. 6(c). CO3 (10)

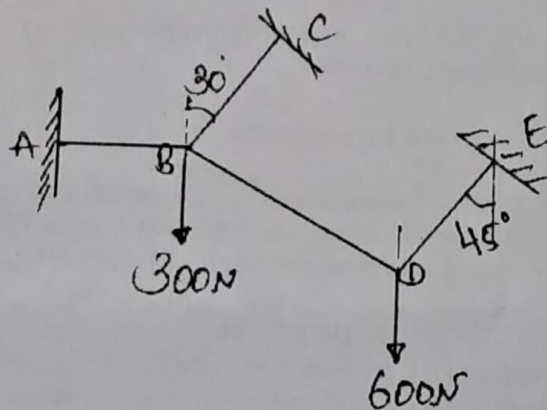


Fig. 6(c)

UNIT- IV

7. a) Determine the centroid of the semi-circle. CO4 (05)
 b) Define the following: (i) Angle of Repose (ii) Angle of Friction. CO4 (05)
 c) Locate the centroid of the shaded area shown in Fig. 7(c). with respect to the axis shown. CO4 (10)

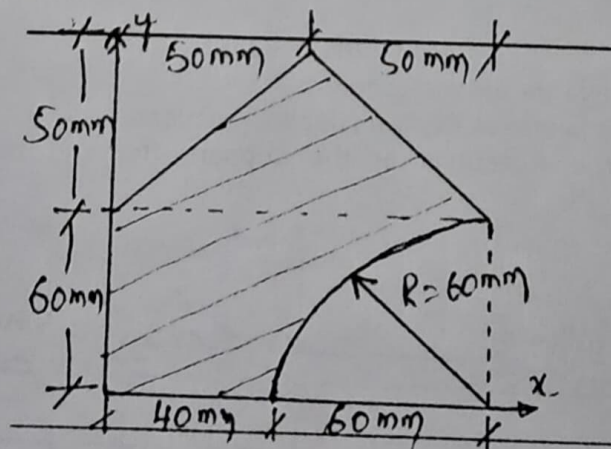


Fig. 7(c).

8. a) Discuss different types of friction. CO4 (05)
 b) What should be the value of θ in Fig. 8(b). Which will make the motion of 1000N block down the plane to impend? The coefficient of friction for all the contact surfaces is $1/3$. CO4 (05)

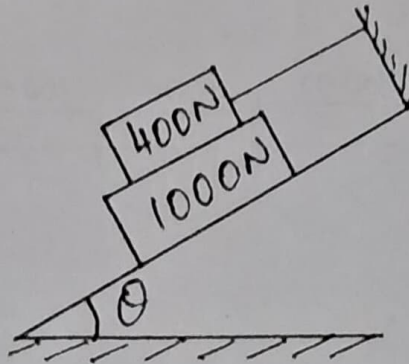


Fig. 8(b).

- c) Locate the centroid of the plane area shown in Fig. 8(c) with respect to the axis shown. CO4 (10)

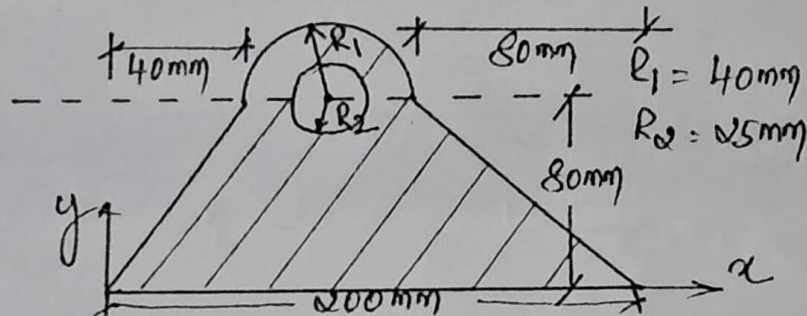


Fig. 8(c).

UNIT - V

9. a) State and prove Perpendicular Axis Theorem. CO5 (05)
 b) Derive an expression to determine the Moment of Inertia for Circle. CO5 (05)
 c) Find the polar Moment of inertia of a plane lamina about point O for the Fig. 9(c). CO5 (10)

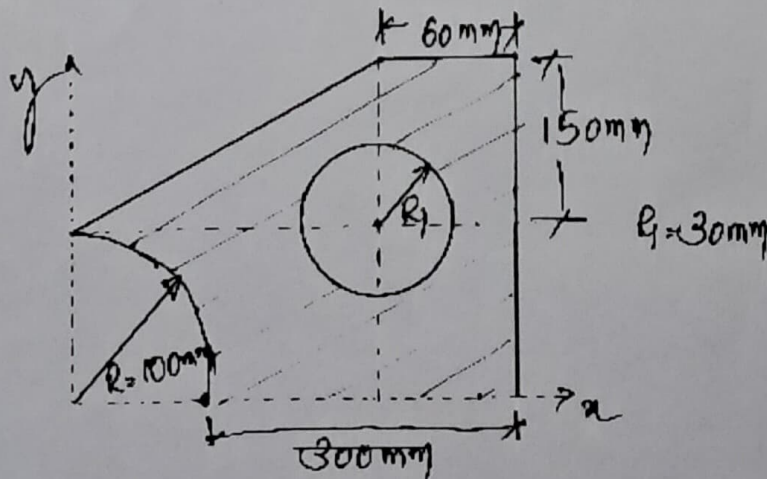


Fig. 9(c)

10. a) State and prove Parallel Axis Theorem. CO5 (05)
 b) Derive an expression to determine the Moment of Inertia for Rectangle. CO5 (05)
 c) For the plane figure shown in Fig. 10(c). Determine the Moment of Inertia and radius of gyration about X- Axis. CO5 (10)

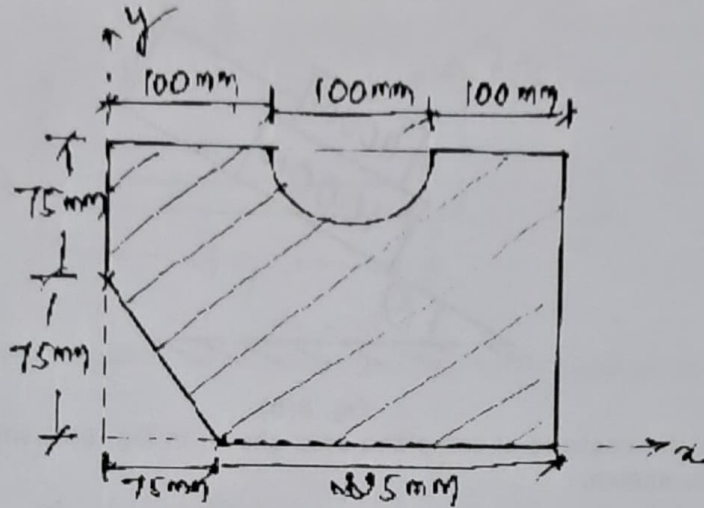


Fig. 10(c)
